

Robotic Microplasty can avoid total knee replacement



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Knee arthritis does not always damage the entire knee. In many patients, wear and tear is limited to just one part of the joint. For such cases, knee microplasty offers an effective solution and can often avoid the need for total knee replacement.

Unlike conventional knee replacement, microplasty changes only the damaged portion of the knee while preserving healthy bone, cartilage, and all major ligaments. By saving the natural

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structures, patients experience better balance, smoother movement, and a more natural-feeling knee.

Although microplasty is a smaller operation, it requires exceptional precision and advanced surgical skill. The implant must be placed with absolute accuracy to restore natural knee movement. This high level of precision was difficult to achieve consistently in the past, limiting the widespread use of the procedure.

Mako Robotic-assisted technology has overcome this challenge. A patient-specific 3-D CT scan is used before surgery to create a detailed plan based on the individual's knee anatomy and natural landmarks. During surgery, the robotic system guides the surgeon to execute this plan accurately, protecting healthy tissues and ensuring precise implant placement.

This robotic precision results in smaller incisions, less tissue damage, minimal blood loss, reduced pain, and faster recovery. Most patients regain mobility earlier and return to daily activities with greater confidence.

The goal of robotic microplasty is a "forgotten joint"—a knee that feels so natural that patients forget they have undergone joint replacement and move freely without constant awareness of an artificial joint.

For the right patients, robotic-assisted partial knee replacement delivers excellent, predictable outcomes while preserving the knee's natural function

We can add these quotes -

1) Why replace the whole knee when only a part is damaged?

2) Microplasty is a smaller operation, but

it demands millimetre-level precision," explains Dr Gaurav Bhardwaj . "Even a small error can affect how the knee feels and moves.

3) The robot doesn't replace the surgeon—it enhances precision and makes results predictable," Dr Bhardwaj adds.

4) The ultimate goal is a "forgotten joint." "A successful knee replacement is one the patient forgets they ever had," says Dr Gaurav Bhardwaj . "It should feel natural, stable, and pain-free in everyday life unrestricted activities

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